MELAKA WATERWAYS & DRAINAGE SYSTEM

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JUDUL: [MELAKA WATERWAYS AND DRAINAGE SYSTEM

SESI PENGAJIAN: [2025 / 2026]

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MELAKA WATERWAYS AND DRAINAGE SYSTEM

KHAIRUL AMRI BIN SHAMSUL ANUAR

This report is submitted in partial fulfillment of the requirements for the Bachelor of Computer Science (Database Management) with Honours.

FACULTY OF INFORMATION AND COMMUNICATION TECHNOLOGY
UNIVERSITI TEKNIKAL MALAYSIA MELAKA

2026

DECLARATION

I hereby declare that this project report entitled

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is written by me and is my own effort and that no part has been plagiarized without citations.

STUDENT : KHAIRUL AMRI BIN SHAMSUL ANUAR_ Date : 22/06/2026

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SUPERVISOR : PN. ROSLEEN BINTI ABD SAMAD__ Date : ________

DEDICATION

[To my beloved parents...]

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Finally, I am thankful to everyone who has contributed directly or indirectly to the success of this project. Your support is deeply appreciated.

ABSTRACT

This report explores the field of urban infrastructure management, focusing specifically on the Melaka Waterways & Drainage System. The primary problem to be solved involves managing public grievances and tracking structural issues effectively, which is addressed through the design of a dedicated drainage complaint system. The research process centered on developing the system design required by the supervisor to establish automated processing logic. Fulfilling the specified academic criteria, this concise study outlines the core system design solutions and operational workflows developed to systematically improve data handling, tracking, and resolution management for waterways within the region.

ABSTRAK

[Abstrak mesti bermula disini. Abstrak mestilah ringkas, DITULIS DALAM SATU LANGKAU dan justifikasi **TIDAK LEBIH DARIPADA 300 PERKATAAN** DALAM SATU MUKASURAT sahaja. Abstrak tidak sama dengan sinopsis atau ringkasan tesis. Abstrak DITULIS DALAM **SATU PERENGGAN**. Ia hendaklah menyatakan dengan ringkas **bidang kajian tesis, masalah yang hendak diselesaikan, cara penyelesaian, proses penyelidikan; dan keputusan yang diperolehi**.]

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LIST OF ABBREVIATIONS

FYP	**-**	**Final Year Project**
MWDS	**-**	**Melaka Waterways and Drainage System**
DFD	**-**	**Data Flow Diagram**
ERD	**-**	**Entity Relationship Diagram**
UI	**-**	**User Interface**

LIST OF ATTACHMENTS

CHAPTER 1: INTRODUCTION

1.1 Introduction

The improper and incompetent management of drainage infrastructure and waterways within Melaka poses a severe threat to public health and safety. Inconsistent maintenance routines routinely lead to debris blockages and prolonged water stagnation.

These neglected systems act as primary breeding grounds for mosquitoes, directly accelerating the transmission of dangerous vector-borne diseases such as dengue virus and malaria within the local community. By failing to systematically maintain these vital assets, current administrative approaches inadvertently foster hazardous environmental conditions that compromise residential well-being.

1.1.1 Subsection 1

Rapid urbanization and climate volatility have placed unprecedented strain on drainage infrastructures, making systematic tracking a necessity rather than an option. Implementing an integrated, data-driven drainage management system bridges the gap between community reports and rapid administrative action, making this project highly relevant to the evolution of smart city infrastructure and modern database administration.

1.1.2 Subsection 2

With the increasing adoption of smart city frameworks and the availability of data-driven municipal technologies, local authorities are transitioning from traditional, reactive workflows to predictive asset management solutions. Many current infrastructure networks lack centralized monitoring interfaces, resulting in delayed responses to blockages and a heightened risk of environmental vector-borne health crises. This project addresses this operational gap by developing an automated

Drainage Complaint and Infrastructure Asset Management System that streamlines community reporting and optimizes maintenance scheduling.

Table 1.1 Understanding the Numbering Levels

Numbering	Name	Role	Example Title
1	Chapter	Main chapter (e.g., Chapter 1)	**Chapter 1: Introduction**
1.1	Subsection	Main subtopic of the chapter	**1.1 Background of the Study**
1.1.1	Sub-subsection	Detailed subtopic under a subsection	**1.1.1 Transportation Issues**
1.1.1.1	Sub-sub-subsection	Further breakdown of a sub-subsection	**1.1.1.1 Unreliable Bus Timing**

1.2 Problem Statement

i. Public Health Risks

Incompetent management of drainage and waterways in Melaka can lead to public health crises. Neglected drainage systems or waterways have been documented to be a common spawn point of mosquitoes, which are capable of spreading diseases such as malaria or the dengue virus. When maintenance is inconsistent, stagnant water and debris blockages become ideal breeding grounds for pests, directly impacting the well-being of the local community. By failing to maintain these systems, the city inadvertently fosters environmental conditions that threaten the health and safety of its residents.

ii. Impact of Dysfunctional Drainage on Flooding Risks

Physical hazards such as structural erosion and blockages reduce flow capacity, especially during extreme weather, like a thunderstorm, increasing the likelihood of property damage due to floods

iii. Infrastructure Degradation Due to Unscheduled Maintenance

The lack of maintenance leads to the gradual degradation of critical drainage and waterway infrastructure. Without a management system to report and locate minor issues in their early stages, they remain undetected until it becomes a significant problem. This ongoing neglect creates physical hazards that can jeopardize public safety. For instance, potential pedestrians walking on the sidewalk, not knowing of the structural instabilities in walkways or adjacent road surfaces.

1.3 Objectives

The system will incorporate interactive data visualization tools to isolate recurring blockages and high-risk flooding zones. By tracking maintenance performance trends, these tools support informed, data-driven decision-making, allowing authorities to prioritize resources and address critical infrastructure vulnerabilities more effectively.

The objective of this study is to

(i) To create a reporting platform that allows users to submit waterway and drainage hazards with GPS location data and photo evidence

(ii) To design a centralized database scheme for storing, categorizing, and tracking reported infrastructure issues to ensure organized data management

(iii) To implement analytics and visualization tools that identify recurring blockages, high-risk flooding zones, and maintenance performance trends to support data-driven decision-making

1.4 Scope of the Project

The system serves as a centralized, responsive web platform designed to streamline municipal infrastructure maintenance across Melaka's local councils, including MBMB, MPHTJ, MPAG, and MPJ. For citizens, the platform provides secure registration, profile updating, and password recovery tools, alongside a dedicated form to file reports on drainage blockages or waterway hazards using precise GPS coordinates and image uploads. Users can track their historical submissions, complete with an auto-generated severity metric, or view ongoing community issues pinned directly across a live interactive map.

Administratively, the application dynamically filters and routes these user-generated reports to the appropriate municipal department based on regional jurisdiction boundaries. Authorized municipal staff and administrators can access a centralized task log to monitor active complaints, update operational tracking statuses, and dispatch specialized maintenance crews directly to identified hazard locations. By structuring data through a relational database backend, the system ultimately converts real-time civic feedback into structured analytics, enabling authorities to easily identify recurring blockages and prioritize critical high-risk flooding zones.

1.4.1 User Scope

The system implements a role-based multi-tier architecture that explicitly defines the capabilities of citizens, municipal staff, and administrators. Citizens interact with a public-facing portal where they can manage their profiles, submit localized drainage or waterway hazard reports with GPS and image data, view their individual complaint histories with auto-generated severity levels, and track public hazard pins on a live interactive map. Municipal staff members handle operational workflows through a centralized task log that lists regional reports filtered by local council jurisdictions (MBMB, MPHTJ, MPAG, or MPJ), with senior staff authorized to update maintenance progress tracking and dispatch physical cleanup crews to site locations. System administrators oversee the complete environment, managing system-wide operational configurations, data security parameters, database integrity checks, and user access credentials to ensure seamless data management across all municipal tiers.

1.4.2 Module Scope

The system is architected around specialized operational modules designed to bridge the gap between civic reporting and municipal response workflows. By isolating core capabilities, the platform maintains clear functional boundaries while ensuring structured data management across all user tiers.

- **Authentication & Profile Management Module:** This foundational module manages identity verification through distinct user roles (Citizen, Staff, and Administrator). It encompasses secure registration, login security parameters, and profile maintenance updates, alongside a verification-driven password recovery mechanism to protect user data integrity.

- **File Report:** Serving as the primary data collection tool, this module enables citizens to submit specific drainage and waterway hazards. The interface captures precise geolocation data, category classifications, textual descriptions, and photo evidence, subsequently issuing a unique, trackable Report ID.

- **View Report:** This interactive module presents historical and real-time report states. Citizens can review their personalized submission logs, including a public-facing live interactive map that displays active, regional hazard pins to keep the community informed.

- **Municipal Dispatch & Task Log Module:** Engineered for administrative workflows, this module automatically routes incoming reports to the corresponding local municipal department (MBMB, MPHTJ, MPAG, or MPJ) based on regional boundaries. Authorized senior staff can access this centralized operational log to evaluate submissions, update remediation progress states, and coordinate the physical dispatch of maintenance crews.

1.5 Expected Outcome

Upon successful completion, this project will deliver a fully functional, web-based Melaka Waterways and Drainage System prototype. The primary tangible deliverable is a web application built on a relational database backend that seamlessly bridges the gap between public reporting and municipal response workflows. Key functioning features will include a secure, multi-tier role-based authentication dashboard, a geospatial-enabled complaint submission portal that captures precise GPS data and photographic evidence, and an interactive public tracking map that displays auto-calculated hazard severity levels. For local councils like MBMB, MPHTJ, MPAG, and MPJ, the system will provide an automated, location-aware task log and dispatch interface to optimize resource deployment.

In terms of non-functional expectations, the final system is designed to meet strict performance and quality benchmarks. The user interface will prioritize intuitive navigation and high responsiveness, ensuring smooth accessibility within a web browser environment for field workers and residents alike.

The database architecture is engineered to achieve optimal query execution speeds, maintaining fast retrieval times for localized mapping coordinates and active task logs during peak traffic hours. Ultimately, the system will achieve a frictionless user experience that minimizes administrative processing delays, turning manual, disconnected public complaints into an organized, data-driven infrastructure maintenance operation.

1.6 Summary

This chapter established the foundational framework for the Melaka Waterways and Drainage System. It introduced the project background by highlighting the environmental and public health risks associated with neglected drainage infrastructure within the Melaka region, followed by a precise problem statement detailing the inefficiencies of current manual tracking workflows. To address these operational gaps, specific, measurable, and actionable project objectives were defined alongside a comprehensive system and user scope encompassing local municipalities like MBMB, MPHTJ, MPAG, and MPJ. Finally, the expected outcomes outlined the functional deliverables and performance goals anticipated upon the successful deployment of the web application prototype.

The next chapter will focus on the Literature Review. It will explore existing theories, methodologies, and technical approaches related to municipal infrastructure tracking and asset management database systems. Additionally, it will analyze and compare similar existing systems to establish a strong theoretical foundation and justify the development choices made for this project

CHAPTER 2: LITERATURE REVIEW AND PROJECT METHODOLOGY

2.1 Literature Review

This section presents a review of previous studies, existing systems, and technologies relevant to the development of the Melaka Waterways and Drainage System. A key benchmark in this review is the *Sistem Pengurusan Aduan Awam* (SISPAA) used by *Jabatan Pengairan dan Saliran* (JPS) Malaysia. As a nationwide, centralized public complaint management framework, SISPAA enables residents to log public infrastructure and environmental concerns directly with federal and state irrigation authorities. While the platform offers a secure, formalized pipeline for routing national complaints, it functions primarily as a text-driven repository that handles general public feedback without specialized local spatial mapping.

Analyzing SISPAA reveals distinct operational limitations in managing regional infrastructure blockages. The platform lacks automated sorting by specific municipal council boundaries, such as MBMB, MPHTJ, MPAG, and MPJ, does not feature real-time, interactive geospatial mapping to pin precise hazard coordinates or dynamically calculate localized severity rankings based on the category of blockages. This project addresses these exact gaps by integrating a visual, map-centric reporting interface and an automated dispatch task log. Developing this prototype provides a localized, responsive solution that directly connects regional reporting with immediate maintenance team dispatch workflows.

2.2 Project Methodology

This section explains the methodology used to develop the Melaka Waterways and Drainage System, including the phases of system development and tools used. Because this project centers entirely on structured data management across multiple regional councils and user roles, the engineering process followed a database-centric lifecycle model. This structured approach ensures that user-reported data routes correctly to regional tables while maintaining optimal query performance.

A diagram of the selected model should be included, along with a brief explanation of each phase (e.g., Planning, Analysis, Design, Implementation, Testing)..

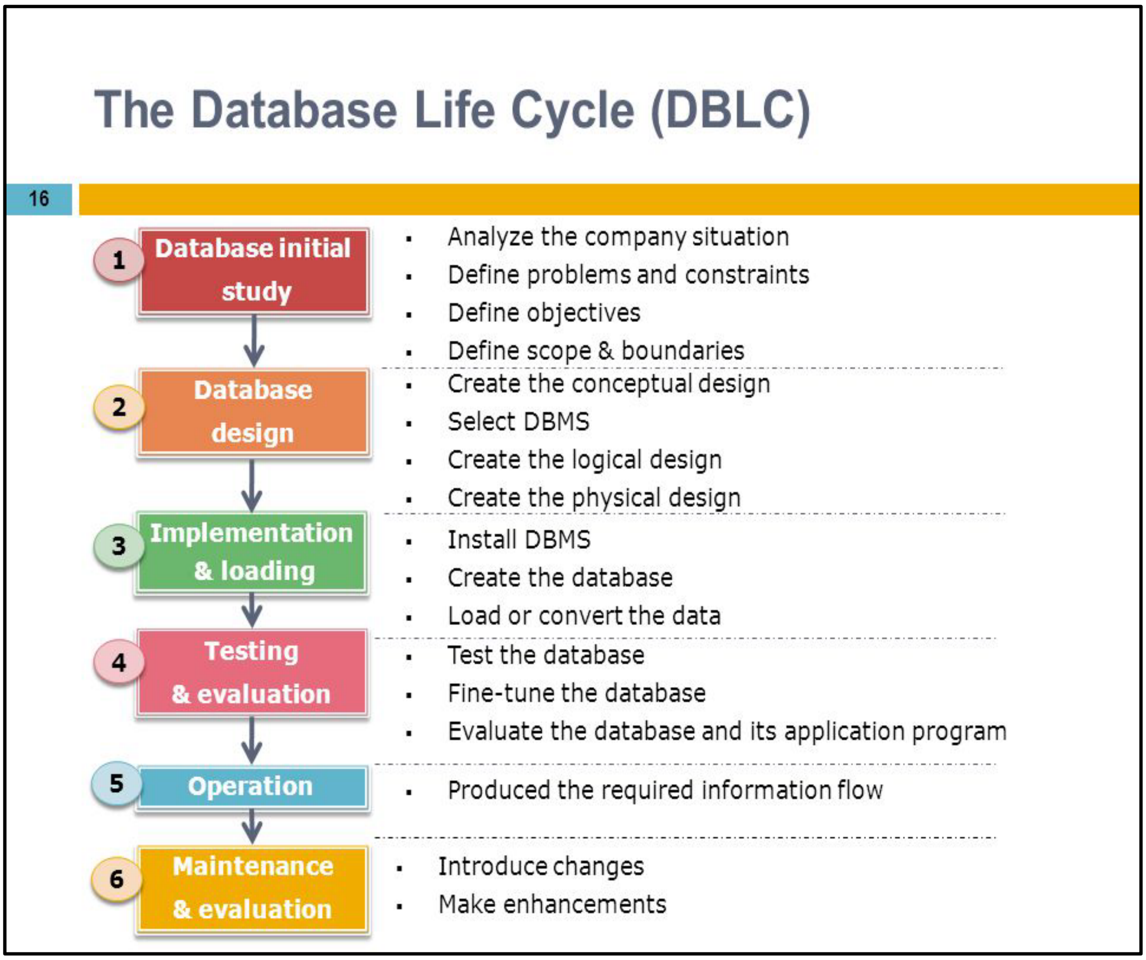

Figure 2.1: Database Life Cycle Stages

The initial lifecycle phases focused on a Planning and Initial Study to map out how complaint data, photographic evidence, and location coordinates would flow from a user's submission to an administrative task log. This explicitly defined the boundaries between public users, municipal staff, and distinct local council jurisdictions like MBMB, MPHTJ, MPAG, and MPJ. Following this, the Analysis and Interface Prototyping phase translated the application forms, buttons, and navigation sidebars into a clear visual blueprint, establishing exactly what data fields, file paths, and coordinate outputs the application would need to capture.

Once the interface prototypes were verified, they were reverse engineered during the Database Design phase to create a logical and physical structure within HeidiSQL, resulting in five interdependent tables: User, Staff, Municipality, Report, and Feedback. During Implementation and Loading, this architecture was deployed on a MariaDB engine using PHP to construct the server-side logic and form validation scripts. Finally, the Testing and Evaluation phase verified the data synchronization between the PHP forms and the MariaDB backend, ensuring that simulated reports, coordinate strings, and image files loaded properly without database execution failures.

2.3 Schedule and Milestones

This section presents the overall timeline of the project, structured into several key development phases from initial inception until final completion. Each phase of the project is planned with specific duration limits to ensure systematic progress, organized milestone tracking, and timely delivery. The core operational phases cover Planning, Analysis, Design, Implementation, Testing, and Documentation. The milestones outlined in Table 2.1 cover the explicit academic deliverables required across both Project I and Project II.

To provide a clearer overview of the project lifecycle, a Gantt chart is included to visualize the precise distribution of tasks across the total development duration. Important project milestones such as the formal proposal submission, frontend prototype development, database backend testing, and the final project presentation are highlighted to serve as critical progress checkpoints. The Gantt chart illustrated in Figure 2.3 maps out these key activities, reflecting the actual learning experience and iterative development steps undertaken throughout the creation of the system.

Table 2.1: Project Milestones

No	Milestone	Description	Expected Completion	Date
1	Problem Identification and Analysis	Identifying the core problem and analyzing requirements	Week 4	7 April 2026
2	Conceptual Design of Proposed System	Designing the overall structure and system framework	Week 6	20 April 2026
3	Physical Design of Proposed System	Developing detailed components, UI, and database design	Week 8	7 May 2026
4	Implementation of Proposed System	System coding, integration, and functional testing	Week 11	25 May 2026

Phase	Sub-Component	Week (Year 2026)																							
		1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**
1. Problem Identification & Analysis	Literature Review																								
	Problem Statement & Objectives																								
2. Conceptual Design of Proposed System	Use Case & Functional Requirement																								
	System Architecture																								
3. Physical Design of Proposed System	Interface Design (GUI)																								
	Database Structure																								
4. Implementation of Proposed System	System Coding																								
	Testing & Debugging																								

Figure 2. 2: Gant Chart of Milestones

2.4 Conclusion

In conclusion, Chapter 2 provides a comprehensive overview of the methodologies used in developing database systems, with a specific emphasis on the Database Life Cycle (DBLC) approach. The six phases of the DBLC are outlined, spanning from the database initial study, database design, implementation and loading, testing and evaluation, operations, to maintenance and evolution. By adopting this structured methodology, the Melaka Waterways and Drainage System ensure a systematic transition from manual reporting methods to a robust, database-driven web environment. The strategic planning of milestones and the project schedule provides a clear framework to ensure that the core data modules are successfully implemented and tested within the established academic timeline.

CHAPTER 3: ANALYSIS

3.1 Introduction

This chapter examines the system analysis phase, focusing on decomposing the platform into core functional sub-modules and gathering empirical operational data. The primary objective is to evaluate current workflows, diagnose systemic inefficiencies, and formulate actionable improvements for the system. This process entails a comprehensive investigation into how infrastructure defects and maintenance tasks are currently managed. By analyzing the flow of operational information, this phase identifies critical bottlenecks encountered by users and develops targeted solutions to optimize their workflows. Ultimately, the analysis concentrates on understanding the complexities of existing drainage and waterway hazard reporting and resolution processes, uncovering operational vulnerabilities, and designing system enhancements that precisely align with stakeholder requirements.

3.2 Problem Analysis

To delve into the shortcomings of the current drainage and waterway hazard reporting process, an unstructured interview and observational study were conducted. Currently, local municipals such as MBMB, MPHTJ, MPAG, and MPJ operate on a manual or fragmented system where infrastructure defects, community complaints, and repair progress are documented using paper formats, telephone logs, or isolated emails. This reliance on physical files and disjointed records creates significant inefficiencies as council staff must manually sift through stacks of paperwork and disconnected files to locate specific maintenance requests. This not only increases the workload for administrative personnel but also consumes valuable time that could be better spent on streamlining field operations and dispatching response crews.

Furthermore, the manual calculation of geographical locations and the lack of precise visual evidence regarding reported blockages or erosion increase the likelihood of human error, potentially resulting in incorrect or flawed prioritization decisions. This lack of automated GPS distance verification, real-time photographic validation, and active severity tracking not only impacts the accuracy of municipal resource allocation but also directly affects the trust, transparency, and safety levels of local residents.

In terms of administrative service, significant delays often occur as staff struggle to manually cross-reference incoming infrastructure reports against shifting repair crew capacities, especially during extreme weather events and peak monsoon seasons when there is a high volume of emergency submissions. This tedious verification period not only affects the efficiency of maintenance delivery but also contributes to public frustration, prolonged flood risks, and structural backlogs within the local council offices.

Additionally, the manual process of updating maintenance statuses, tracking resolving tasks, and generating handwritten or manually typed summaries for operational performance reports adds further complexity and time consumption to the workflow. Administrative personnel must manually update and track each complaint's status, which can be prone to oversight, lost data, and communication delays. Manually compiled asset and hazard summaries are not only time-consuming to produce but also highly prone to errors in data entry and regional trend interpretation.

3.3 Proposed System

The proposed Melaka Waterways & Drainage System (MWDS) is a web-based platform designed to transition municipal hazard reporting from outdated manual methods to a centralized, database-driven workflow. Developed using a PHP backend architecture deployed on a MariaDB engine, the new system directly resolves communication delays and resource allocation inefficiencies by automating the collection and tracking of infrastructure data. Key features include real-time public reporting with visual documentation and precise GPS tagging, a centralized dashboard for municipal staff to manage tasks, and administrative analytical tools that process citizen feedback trends to justify future funding requests.

Interaction with the platform is structured across three distinct user roles. Public citizens register accounts to submit waterway or drainage reports and log post-repair evaluation feedback; field and senior staff log in to validate incoming submissions, assign tasks, and update structural repair statuses; and administrative authorities oversee staff accounts, analyze negative feedback loops, and strategically coordinate resource allocation across distinct local council jurisdictions.

3.4 User Requirements

This section lists the functional and non-functional requirements of the system. To ensure a fully functional system, all of these requirements have to be met with no compromise.

Table 3.1: Functional Requirements of the System

No	Functional Requirement	Description
FR01	Citizen creates Account	System should allow user to register a new account
FR02	Citizen Login	System shall allow citizens to log in to their dashboard.
RF03	Forget Password	System shall allow users to reset their account passwords.
FR04	Update Profile	System shall allow citizens to modify their profile information.
FR05	File a Report	System shall allow citizens to submit drainage hazard reports with photos and GPS.
FR06	Submit Resolved Report Feedback	System shall allow citizens to submit satisfaction feedback on resolved cases.
FR07	Staff Login	System should allow municipal staff members to log in.
FR08	View Centralized Task Log (Staff)	System shall allow staff to view and filter incoming complaints.
FR09	Manage Reports & Status Control	System shall allow senior staff to update report statuses.
FR10	Dispatch Maintenance Teams	System shall allow senior staff to assign repair crews to coordinates.
FR11	Provide On-Site Task Updates	System shall allow field crews to submit progress updates and photos.
FR12	Admin Login	System shall allow administrators to log in to the system.

FR13	Staff Management	System shall allow administrators to perform CRUD operations on staff profiles.
FR14	View Centralized Task Log (Admin)	System shall allow administrators to view all historic and active records.
FR15	Manage Reports & Dispatch Teams	System shall allow administrators to update statuses and assign field crews.
FR16	View Feedback Analytics & Reporting	System shall allow administrators to view satisfaction trends and complaint heatmaps.

Table 3.2: Non-Functional Requirements Example

No	Non-Functional Requirement	Description
NFR01	System Availability	The system should be available 24/7.
NFR02	Data Encryption	System shall encrypt all user passwords using secure hashing algorithms before database persistence.
NFR03	Access Control	System shall enforce strict Role-Based Access Control (RBAC) to ensure users can only access modules authorized for their roles.
NFR04	System Performance	System shall process and render database queries, spatial tasks, and log filtering actions within 3 seconds.
NFR05	File Upload Capacity	System shall support concurrent multipart image uploads up to 5 images per report.
NFR06	Interface Usability	System shall feature a responsive user interface with clean navigation menus optimized for both mobile and desktop screens.

3.5 Data Flow Diagram (DFD)

This section presents the context diagram and Level 1 data flow diagram (DFD) to illustrate the logical process flow of the proposed system. The diagrams highlight how information moves seamlessly between external entities, system processes, and data stores, ensuring that inputs are transformed into meaningful outputs through defined automation steps.

3.5.1 Context Diagram

The system is represented as a single overarching process that interacts directly with external entities such as citizens, municipal staff, and administrators. Data flows into the system when citizens submit complaints, which are then acknowledged and routed back as confirmation or updates. Municipal staff provide operational input and receive task assignments, while administrators oversee system outputs such as performance reports. This high-level view emphasizes the boundaries of the system, showing how information enters, is processed, and exits without detailing internal mechanisms, thereby establishing a clear logical framework for understanding the system's role in managing drainage complaints.

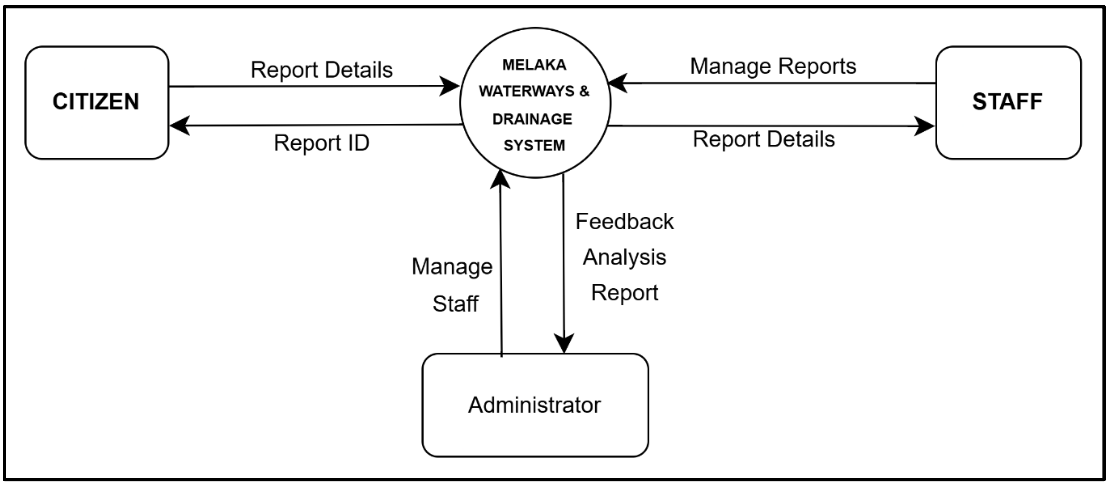

Figure 2. 3: Context Diagram of MWDS

3.5.2 Level 0 DFD

At Level 0, the system is decomposed into its core sub-processes, showing how data flows internally between complaint submission, classification, assignment, tracking, and resolution. Citizens initiate the process by submitting complaints, which are validated and stored in the complaint database. Municipal staff then access these records to classify and assign tasks, while administrators monitor progress through performance reports generated from task logs. Each process interacts with relevant data stores to ensure information is consistently updated and retrievable, while external entities receive timely feedback such as acknowledgments, updates, or resolutions. This structured breakdown highlights the logical automation of workflows, making the system's operations transparent and efficient.

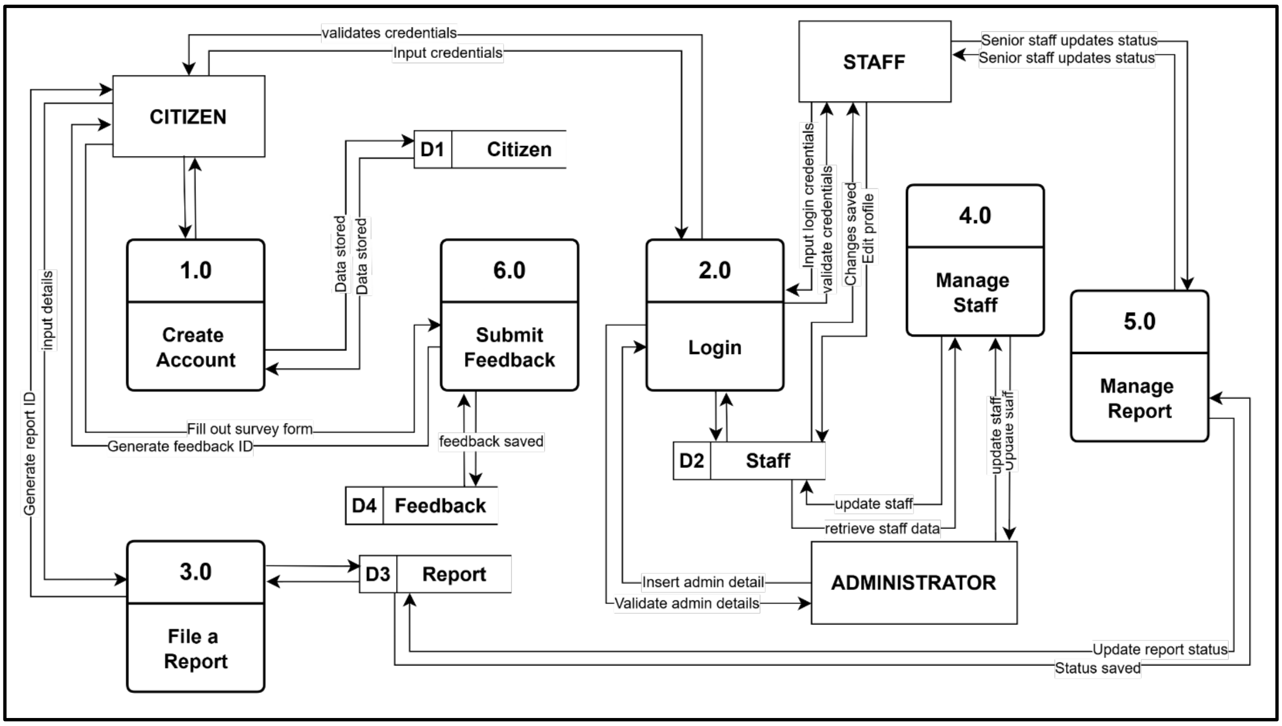

Figure 2.4 DFD Level 0 of MWDS

3.6 Summary

The chapter identifies several requirements for the proposed drainage complaint system, focusing on efficient data handling, transparency, and responsiveness. Functionally, the system must allow citizens to submit complaints with supporting details, ensure validation and classification of reports, and provide municipal staff with tools to assign and track tasks. Non-functional requirements emphasize security in handling sensitive user data, performance in processing multiple complaints simultaneously, and reliability in delivering timely updates. To address these needs, the system integrates a centralized complaint database, automated workflows for classification and resolution, and feedback mechanisms that notify users of progress. Together, these features demonstrate how the system logically transforms raw input into actionable outcomes, meeting both user expectations and organizational goals.

CHAPTER 4: DESIGN

4.1 System Architecture

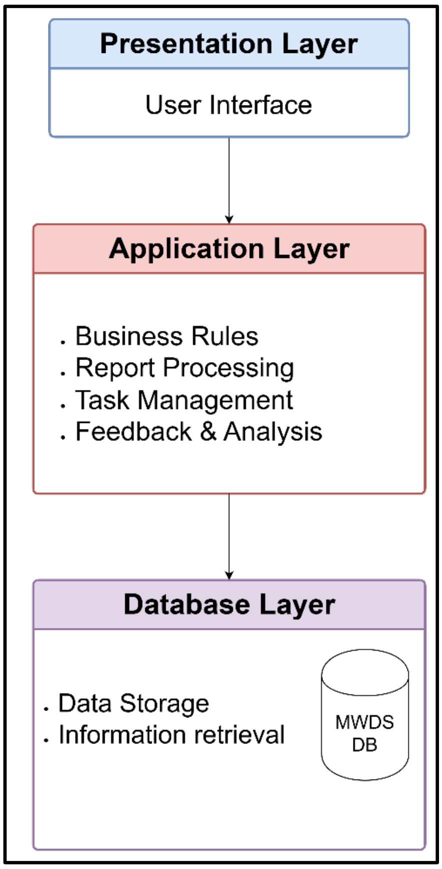

Figure 2. 5: System Architecture of MWDS

This section presents the **System Architecture Diagram** to visually illustrate the major components of the proposed system and their interactions. The architecture adopts a three-tier structure, consisting of the Presentation Layer, Application Layer, and Database Layer. Each layer plays a distinct role in ensuring modularity, scalability, and efficient data flow.

4.2 Data Model Design

A data model serves as the conceptual framework that defines how data is structured, stored, and accessed within the system. It provides a logical representation of entities, their attributes, and the relationships between them. By organizing data in this way, the model ensures consistency, supports efficient queries, and facilitates integration across system components.

For this project, the data model is represented using a normalized Entity-Relationship Diagram (ERD). The ERD captures the essential entities of the system, such as User, Staff, Report, Feedback, Notification, and Database Logs, along with their attributes and interconnections. Normalization is applied to reduce redundancy and maintain data integrity, ensuring that each entity is well-defined and supports scalable operations.

The diagram (see Figure 4.2) illustrates:

- **Entities:** Core tables that represent system participants and processes.
- **Attributes:** Key data fields stored within each entity.
- **Relationships:** Logical connections (1–1, 1–M, M–M) that define how data flows between entities.

This structured approach highlights how the system's information is managed and accessed, forming the backbone of the overall architecture.

4.2.1 Conceptual Design

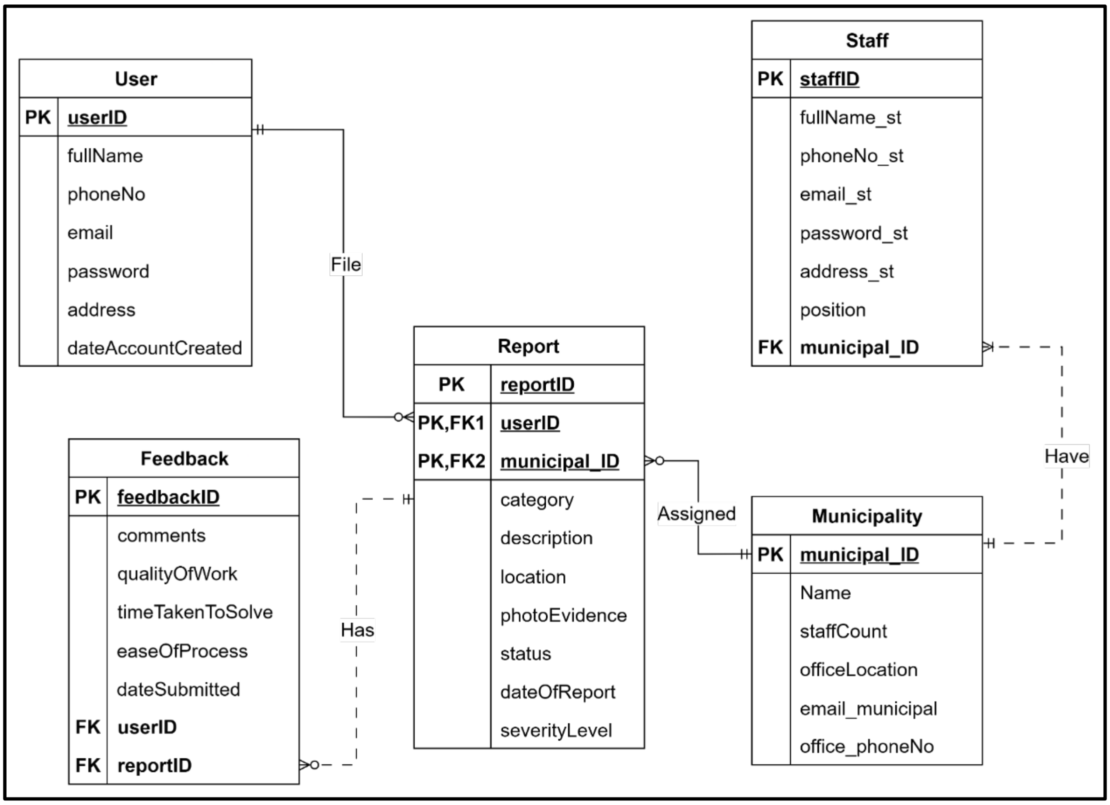

Figure 2. 6: Entity Relationship Diagram of MWDS

4.2.2 Business Rules

The business rules define how entities interact and ensure data consistency across the system. In this system:

1. Users can file multiple reports, each report must belong to one user only.
2. Each Staff are assigned to one municipality, but each municipality can have multiple staff members.
3. Each report must be from only one municipal, but each municipal can have multiple reports.
4. Each feedback entry must correspond to a single report.

4.2.3 Logical Design

This section presents the Data Dictionary for the drainage complaint management system, developed from the data model in Section 4.2. The Data

Dictionary provides a detailed definition of the database structure by listing all entities, attributes, data types, lengths, keys, default values, and constraints. It serves as a blueprint for implementation, ensuring clarity, consistency, and accuracy across the development team. By documenting each table systematically, the Data Dictionary supports communication among stakeholders, reduces design errors, and strengthens data integrity. Tables 4.1 to 4.4 highlight the major entities of the system, including User, Staff, Municipality, Report, and Feedback, along with their attributes and relationships. This structured documentation enhances maintainability, supports future system upgrades, and ensures that the database design aligns with the functional and non-functional requirements of the project.

Table 4.1: Table User

USER									
ATTRIBUTE NAME	**CONTENTS**	**DATA TYPE AND SIZE**	**FORMAT**	**RANGE**	**REQUIRED?**	**DEFAULT VALUE**	**UNIQUE?**	**PK OR FK**	**FK REFERENCED TABLE**
userID	Unique identifier for user	INT(11)	Auto Increment	≥1	Yes		Yes	PK	
fullName	Full name of user	VARCHAR(255)	Text		Yes				
phoneNo	Contact number	VARCHAR(20)	Digits		Yes				
email	Email address	VARCHAR(255)	user@domain		Yes		Yes		
password	Account password	VARCHAR(255)	Encrypted		Yes				
address	Residential address	TEXT			Yes				
dateAccountCreated	Account creation date	DATETIME	YYYY-MM-DD HH:MM		Yes	CURRENT_ TIMESTAMP			

Table 4.2: Table Staff

STAFF									
ATTRIBUTE NAME	**CONTENTS**	**DATA TYPE AND SIZE**	**FORMAT**	**RANGE**	**REQUIRED?**	**DEFAULT VALUE**	**UNIQUE?**	**PK OR FK**	**FK REFERENCED TABLE**
staffID	Unique staff identifier	VARCHAR(10)	Alphanumeric		Yes		Yes	PK	
fullName_st	Staff full name	VARCHAR(255)	Text		Yes				
phoneNo_st	Staff contact number	VARCHAR(20)	Digits						
email_st	Staff email	VARCHAR(255)	staff@domain		Yes		Yes		
password_st	Staff password	VARCHAR(255)	Encrypted		Yes				
address_st	Staff address	TEXT			Yes				
position	Staff role/position	VARCHAR(100)	Text		Yes				
municipal_ID	Municipality assigned	INT(11)			Yes		Yes	FK	MUNICIPALITY

Table 4.3: Table Municipality

MUNICIPALITY									
ATTRIBUTE NAME	**CONTENTS**	**DATA TYPE AND SIZE**	**FORMAT**	**RANGE**	**REQUIRED?**	**DEFAULT VALUE**	**UNIQUE?**	**PK OR FK**	**FK REFERENCED TABLE**
municipal_ID	Unique municipality ID	INT(11)	Auto Increment	≥1	Yes		Yes	PK	
Name	Municipality name	VARCHAR(255)	Text		Yes				
staffCount	Number of staff	INT(11)	Numeric		Yes				
officeLocation	Office location	VARCHAR(255)	Text		Yes				
email_municipal	Municipality email	VARCHAR(255)	municipal@domain		Yes		Yes		
office_phoneNo	Office phone number	INT(11)	Digits		Yes				

Table 4.4: Table Report

REPORT									
ATTRIBUTE NAME	**CONTENTS**	**DATA TYPE AND SIZE**	**FORMAT**	**RANGE**	**REQUIRED?**	**DEFAULT VALUE**	**UNIQUE?**	**PK OR FK**	**FK REFERENCED TABLE**
reportID	Unique report ID	INT(11)	Auto Increment	≥1	Yes		Yes	PK	
userID	Linked user ID	INT(11)					Yes	FK	USER
municipal_ID	Linked municipality ID	INT(11)			Yes		Yes	FK	MUNICIPALITY
category	Complaint category	VARCHAR(100)	Text		Yes				
description	Complaint description	TEXT			Yes				
location	Complaint location	VARCHAR(255)			Yes				
latitude	Latitude coordinate	DECIMAL(10,8)			Yes				
longitude	Longitude coordinate	DECIMAL(11,8)			Yes				
photoEvidence	Photo evidence path	VARCHAR(255)	File path		Yes				
status	Complaint status	VARCHAR(50)	Text		Yes	Pending			

dateOfReport	Date of complaint	DATETIME	YYYY-MM-DD HH:MM		Yes	CURRENT_ TIMESTAMP			
severityLevel	Severity level	VARCHAR(50)	Text	High/ Medium/ Low	Yes				

Table 4.5: Table Feedback

FEEDBACK									
ATTRIBUTE NAME	**CONTENTS**	**DATA TYPE AND SIZE**	**FORMAT**	**RANGE**	**REQUIRED?**	**DEFAULT VALUE**	**UNIQUE?**	**PK OR FK**	**FK REFERENCED TABLE**
feedbackID	Unique feedback ID	INT(11)	Auto Increment		Yes		Yes	PK	
comments	Feedback comments	TEXT			No				
qualityOfWork	Rating of work quality	VARCHAR(100)	Text		Yes				
timeTakenToSolve	Resolution time feedback	VARCHAR(100)	Text		Yes				
easeOfProcess	Process satisfaction	VARCHAR(100)	Text		Yes				
dateSubmitted	Feedback submission date	DATETIME	YYYY-MM-DD HH:MM		Yes	CURRENT_ TIMESTAMP			
reportID	Linked report ID	INT(11)			Yes		Yes	FK	REPORT

4.2.4 Physical Design

The physical design phase translates the logical database model into an actual database structure implemented in MariaDB/MySQL. This stage defines how data is stored, accessed, indexed, and optimized for performance in the deployed drainage complaint management system.

The design ensures that entities such as User, Staff, Municipality, Report, and Feedback are implemented with appropriate data types, constraints, and relationships to maintain integrity and efficiency. Foreign key references enforce consistency across tables, while triggers automate business rules such as severity classification and municipality assignment.

Indexes on frequently queried fields (e.g., userID, municipal_ID, reportID) improve retrieval speed, and default values (e.g., status = Pending, dateAccountCreated = CURRENT_TIMESTAMP) streamline data entry. This section includes the complete Data Definition Language (DDL) scripts of the system, serving as the blueprint for database deployment, documentation, and future maintenance.

```
CREATE TABLE IF NOT EXISTS `feedback` (
  `feedbackID` int(11) NOT NULL AUTO_INCREMENT,
  `comments`   text DEFAULT NULL,
  `qualityOfWork` varchar(100) DEFAULT NULL,
  `timeTakenToSolve` varchar(100) DEFAULT NULL,
  `easeOfProcess` varchar(100) DEFAULT NULL,
  `dateSubmitted` datetime DEFAULT current_timestamp(),
  `userID` int(11) DEFAULT NULL,
  `reportID` int(11) DEFAULT NULL,
  PRIMARY KEY (`feedbackID`),
  KEY `userID` (`userID`),
  KEY `reportID` (`reportID`),
  CONSTRAINT `feedback_ibfk_1` FOREIGN KEY (`userID`)
REFERENCES `user` (`userID`) ON DELETE CASCADE ON UPDATE
CASCADE,
  CONSTRAINT `feedback_ibfk_2` FOREIGN KEY (`reportID`)
REFERENCES `report` (`reportID`) ON DELETE CASCADE ON UPDATE
CASCADE
);
```

Figure 2. 7 Table FEEDBACK

```
CREATE TABLE IF NOT EXISTS `municipality` (
  `municipal_ID` int(11) NOT NULL AUTO_INCREMENT,
  `Name` varchar(255) NOT NULL,
  `staffCount` int(11) NOT NULL,
  `officeLocation` varchar(255) NOT NULL,
  `email_municipal` varchar(255) NOT NULL,
  `office_phoneNo` int(11) NOT NULL,
  PRIMARY KEY (`municipal_ID`)
);
```

Figure 2. 8 Table MUNICIPALITY

```
CREATE TABLE IF NOT EXISTS `report` (
  `reportID` int(11) NOT NULL AUTO_INCREMENT,
  `userID` int(11) NOT NULL,
  `municipal_ID` int(11) NOT NULL,
  `category` varchar(100) NOT NULL,
  `description` text NOT NULL,
  `location` varchar(255) NOT NULL,
  `latitude` decimal(10,8) DEFAULT NULL,
  `longitude` decimal(11,8) DEFAULT NULL,
  `photoEvidence` varchar(255) DEFAULT NULL,
  `status` varchar(50) DEFAULT 'Pending',
  `dateOfReport` datetime DEFAULT current_timestamp(),
  `severityLevel` varchar(50) NOT NULL,
  PRIMARY KEY (`reportID`),
  KEY `userID` (`userID`),
  KEY `report_ibfk_2` (`municipal_ID`),
  CONSTRAINT `report_ibfk_1` FOREIGN KEY (`userID`)
REFERENCES `user` (`userID`) ON DELETE CASCADE ON UPDATE
CASCADE,
  CONSTRAINT `report_ibfk_2` FOREIGN KEY
(`municipal_ID`) REFERENCES `municipality`
(`municipal_ID`) ON UPDATE CASCADE
);
```

Figure 2. 9 Table REPORT

```
CREATE TABLE IF NOT EXISTS `staff` (
  `staffID` varchar(10) NOT NULL,
  `fullName_st` varchar(255) NOT NULL,
  `phoneNo_st` varchar(20) DEFAULT NULL,
  `email_st` varchar(255) NOT NULL,
  `password_st` varchar(255) NOT NULL,
  `address_st` text DEFAULT NULL,
  `position` varchar(100) DEFAULT NULL,
  `municipal_ID` int(11) DEFAULT NULL,
  PRIMARY KEY (`staffID`),
  UNIQUE KEY `email_st` (`email_st`),
  KEY `fk_staff_municipality` (`municipal_ID`),
  CONSTRAINT `fk_staff_municipality` FOREIGN KEY
(`municipal_ID`) REFERENCES `municipality`
(`municipal_ID`) ON DELETE SET NULL ON UPDATE CASCADE
);
```

Figure 2. 10 Table STAFF

```
CREATE TABLE IF NOT EXISTS `user` (
  `userID` int(11) NOT NULL AUTO_INCREMENT,
  `fullName` varchar(255) NOT NULL,
  `phoneNo` varchar(20) DEFAULT NULL,
  `email` varchar(255) NOT NULL,
  `password` varchar(255) NOT NULL,
  `address` text DEFAULT NULL,
  `dateAccountCreated` datetime DEFAULT
current_timestamp(),
  PRIMARY KEY (`userID`),
  UNIQUE KEY `email` (`email`)
);
```

Figure 2. 11 Table USER

Table 4.5: List of Database Objects using MariaDB Command

No.	Goal	Collections Involved	MongoDB Command	Type	Description
1	Automatically classify severity and assign municipality when a report is inserted	Report, Municipality	`CREATE TRIGGER before_report_insert BEFORE INSERT ON report FOR EACH ROW BEGIN … END;`	Trigger	Cleans category input, sets severity level, and maps location/postcode to the correct municipality before saving a new report.
2	Improve search speed for reports by category	Report	`CREATE INDEX idx_report_category ON report(category);`	Index	Speeds up queries filtering reports by category, useful for staff dashboards and analytics.

4.3 Graphical User Interface Design

The user interface is designed to be as intuitive as possible to make sure that all types of users can navigate through the system without having to figure anything out.

- Navigation System:

Citizen:

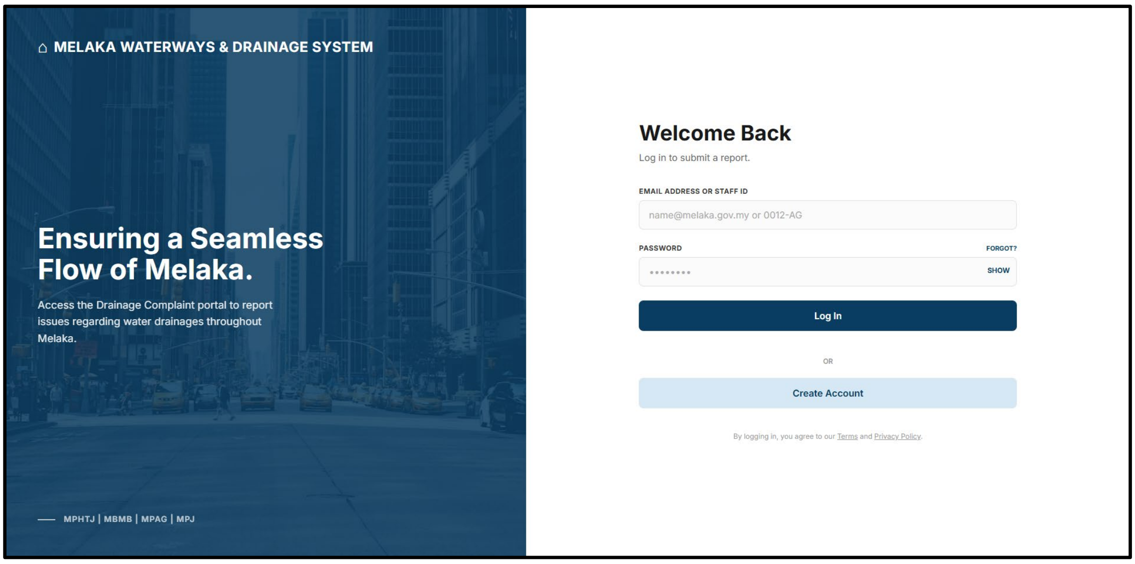

Figure 4.3.1 Login Page

The figure above is the login page for the system. The login page is accessible for all tiers of users for this system. Citizens, staff members, and administrators will use the same login page. Citizens can create an account, and login using their credentials. If they happen to forget their password, they can click the "FORGOT?" hyperlink and reset it from there.

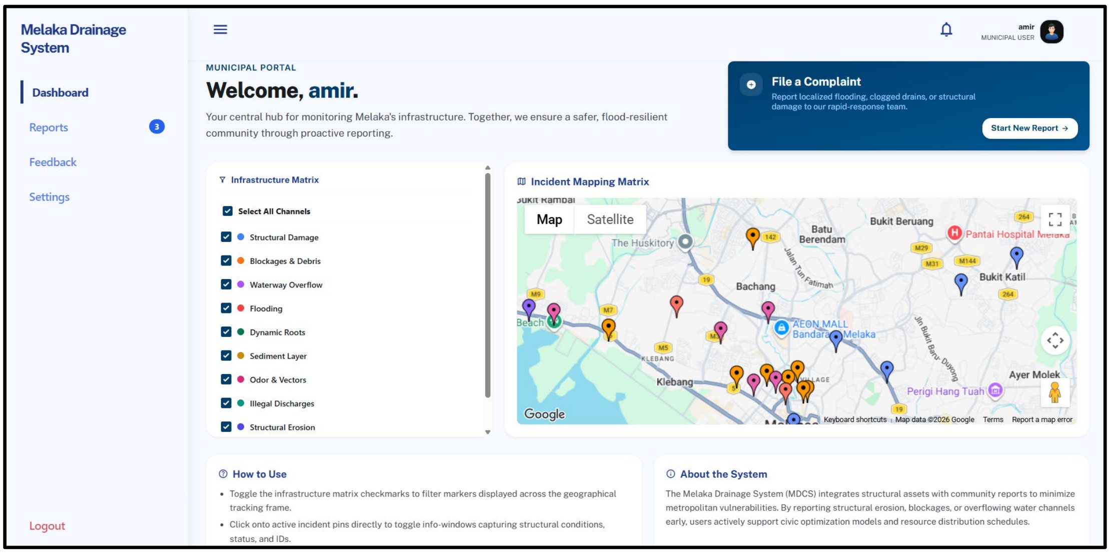

Figure 4.3.2: Citizen Dashboard

After entering credentials, citizen will be able to access their homepage. There, they can view the live-report map, where it shows the currently active reports within Melaka.

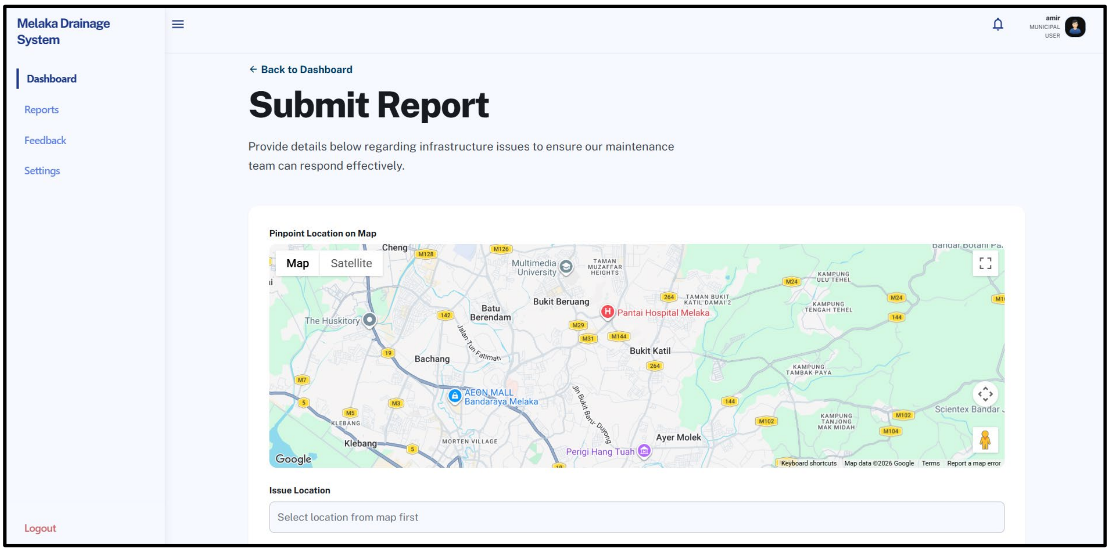

Figure 4.3.3: File a Report

From the homepage, after clicking "File a Report" button, Citizen will be navigated to the Report Form page.

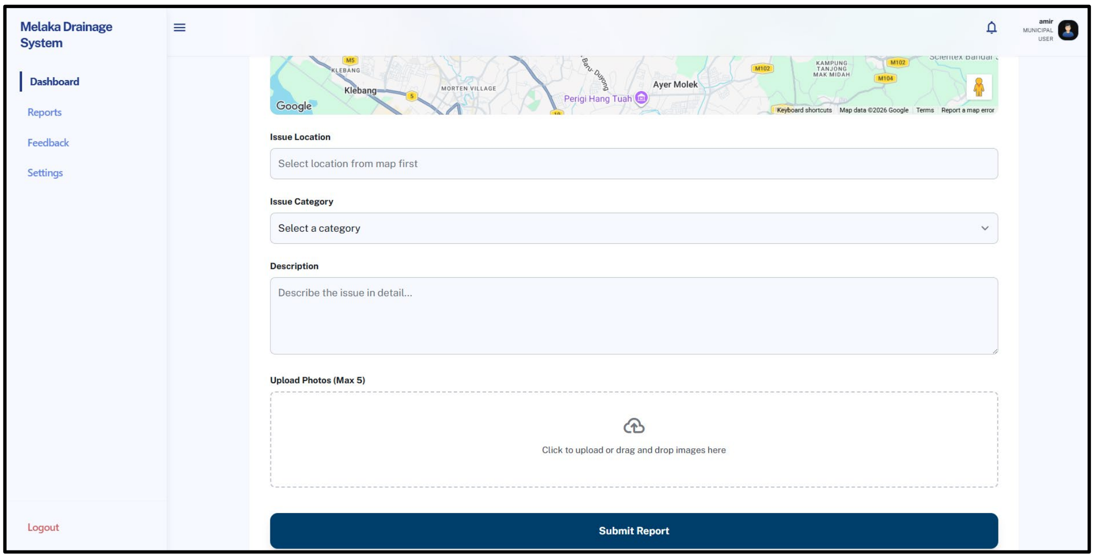

Figure 4.3.4 Submit Report

To complete, there are 4 steps to submit a report. Starting with choosing a location from the map shown. Next is choosing a category of report, followed by a description textbox. The final step is an image submission hub.

Staff:

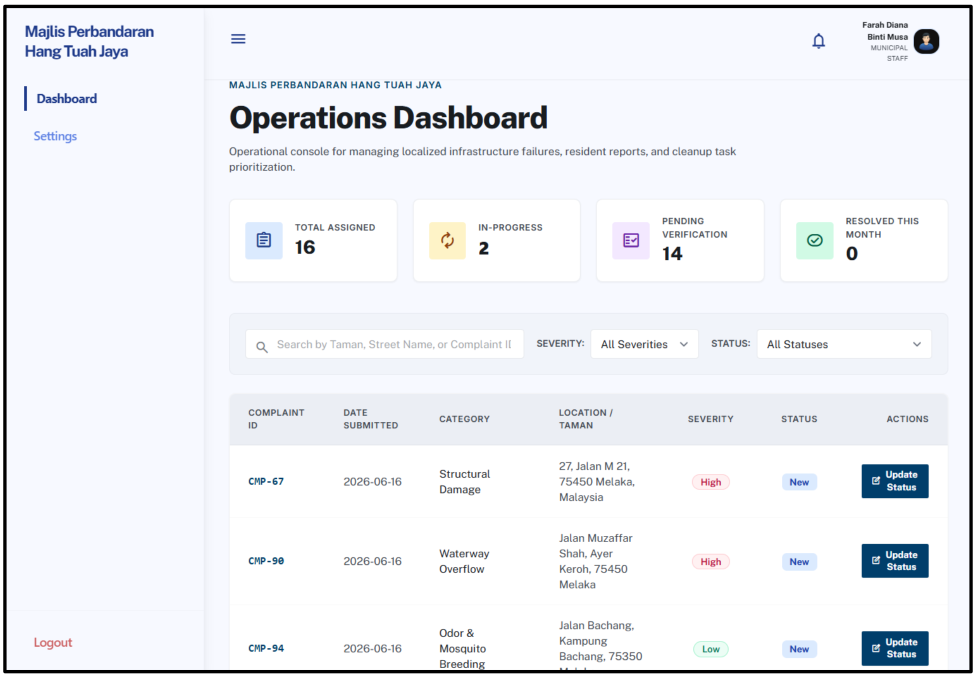

Figure 4.3.5 Staff Dashboard

Upon login, staff members will be able to see this dashboard. Here, they can view the details of the report, to prepare before being dispatched.

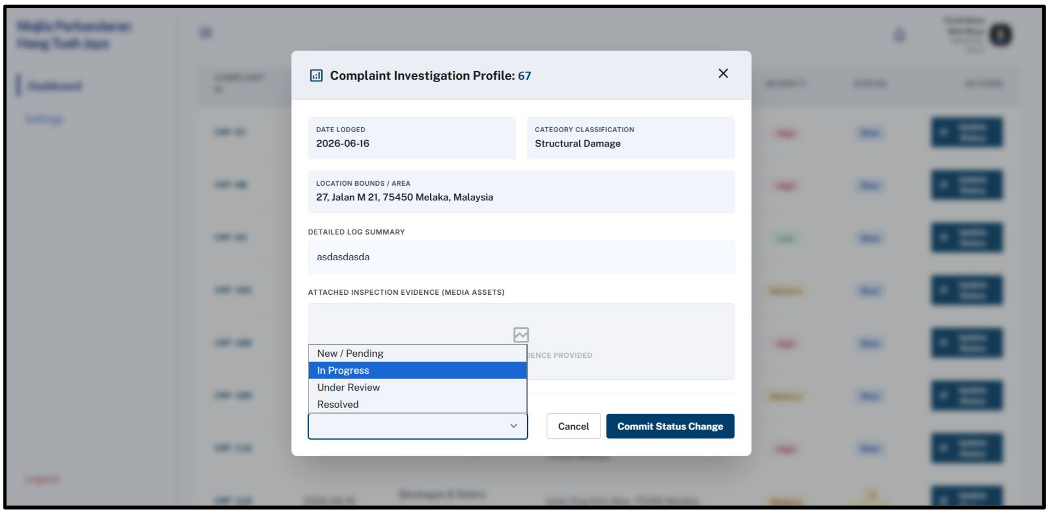

Figure 4.3.6 Report Status Update

Admins and Senior Manager positions have the authority to change the status of a report. Status reports will be changed after every update from the dispatch team.

Admin

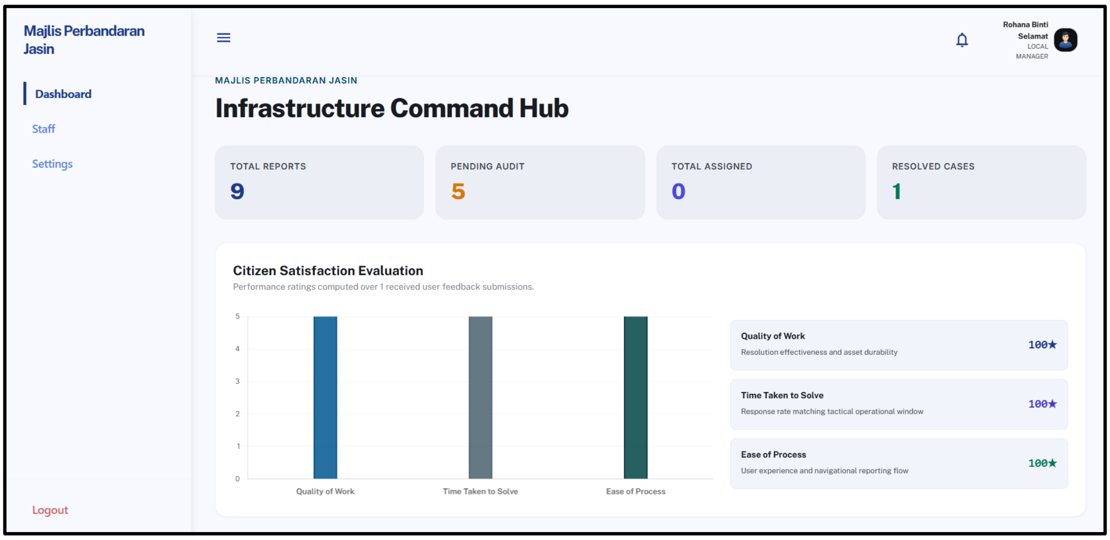

Figure 4.3.7 Admin Dashboard

Admins will see this dashboard upon logging in. There are analytics from Citizen feedback, showing the average report resolution scores.

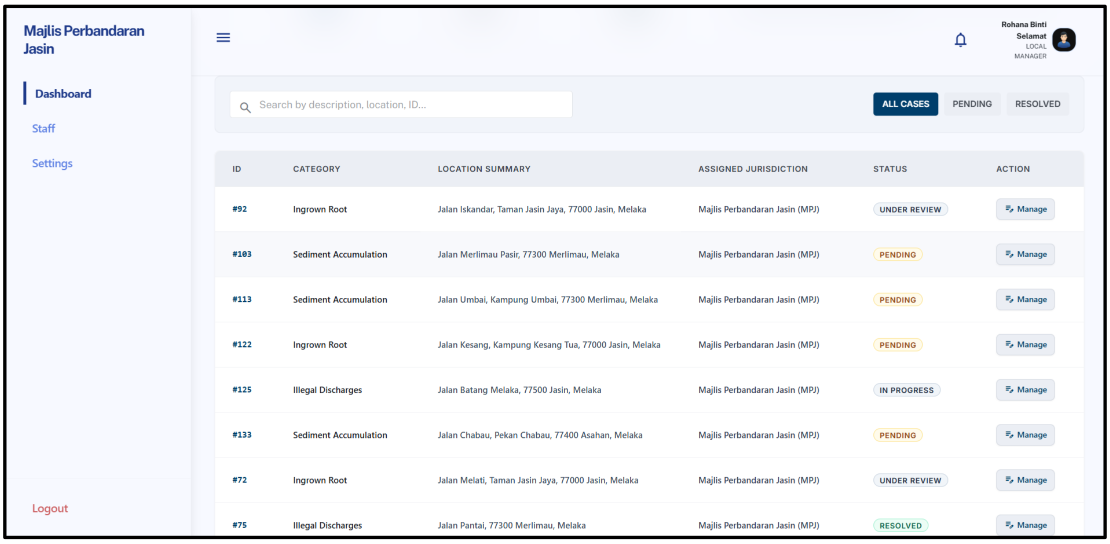

Figure 4.3.8 Admin Task Log

Scrolling down below the analytics, admin can see the list of all reports, in their municipality. Admin has authority to change status as well.

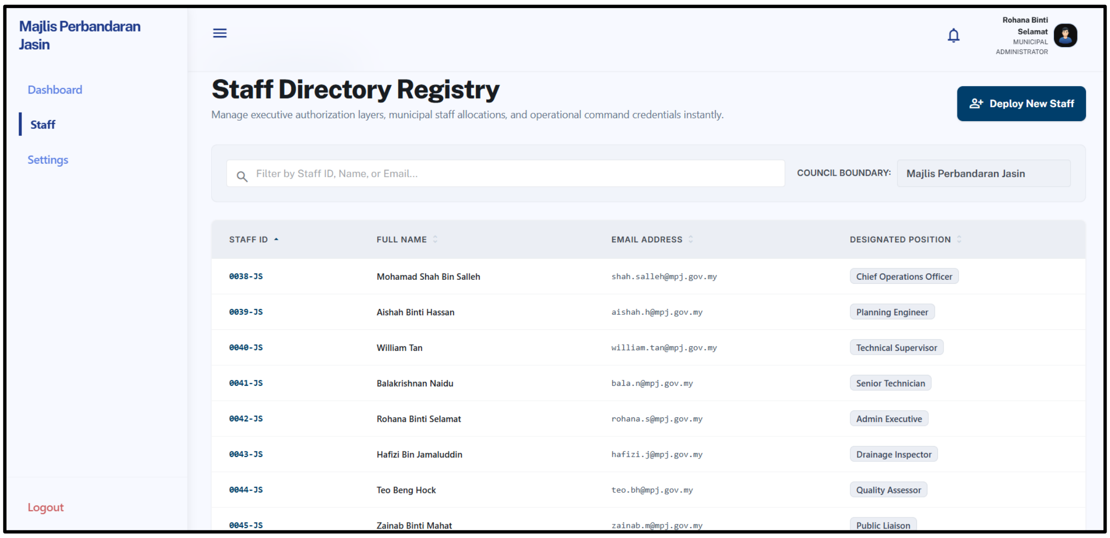

Figure 4.3.9 Admin Staff Directory

The staff directory is only available to the admin. Here, admin can add, update, or delete staff as they please.

- Input Design: Show forms/screens where users enter data.

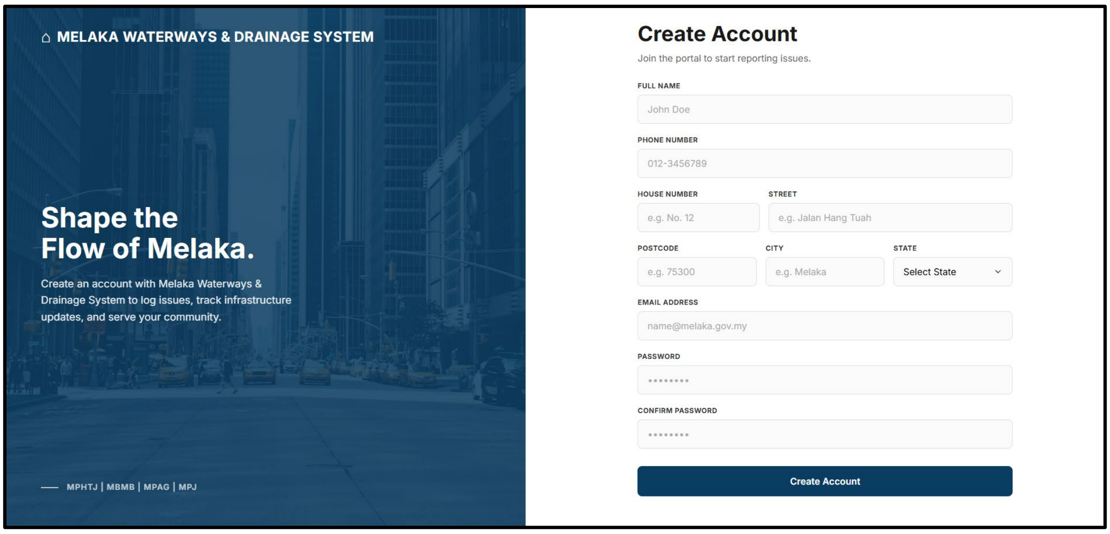

Figure 4.3.10 Create Account

This is the page for creating an account. This page can only be used for Citizen-tier access. Staff members creating an account would result in a citizen account.

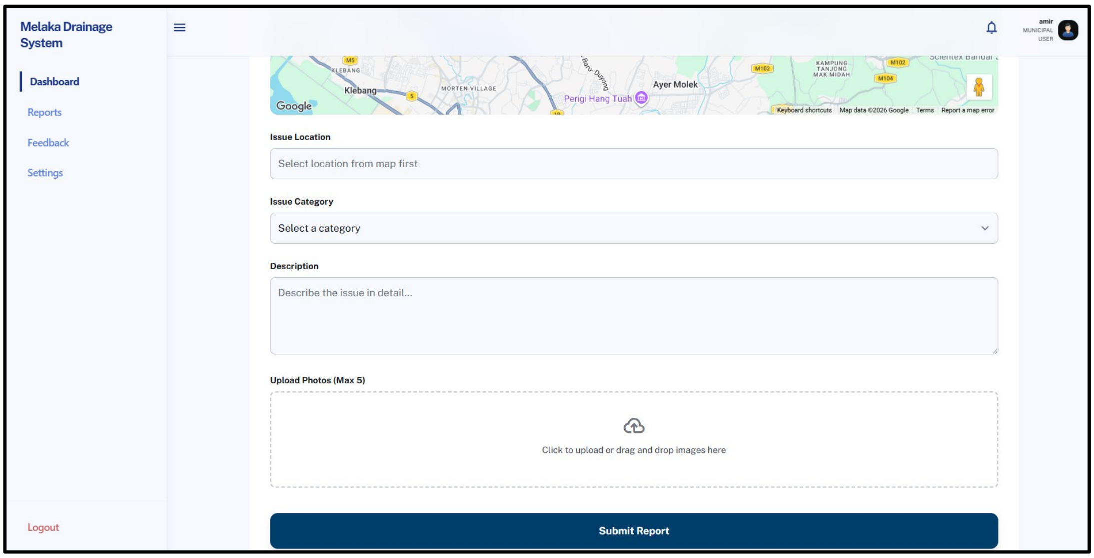

Figure 4.3.11 Fill in Report Form

To complete, there are 4 steps to submit a report. Starting with choosing a location from the map shown. Next is choosing a category of report, followed by a description textbox. The final step is an image submission hub.

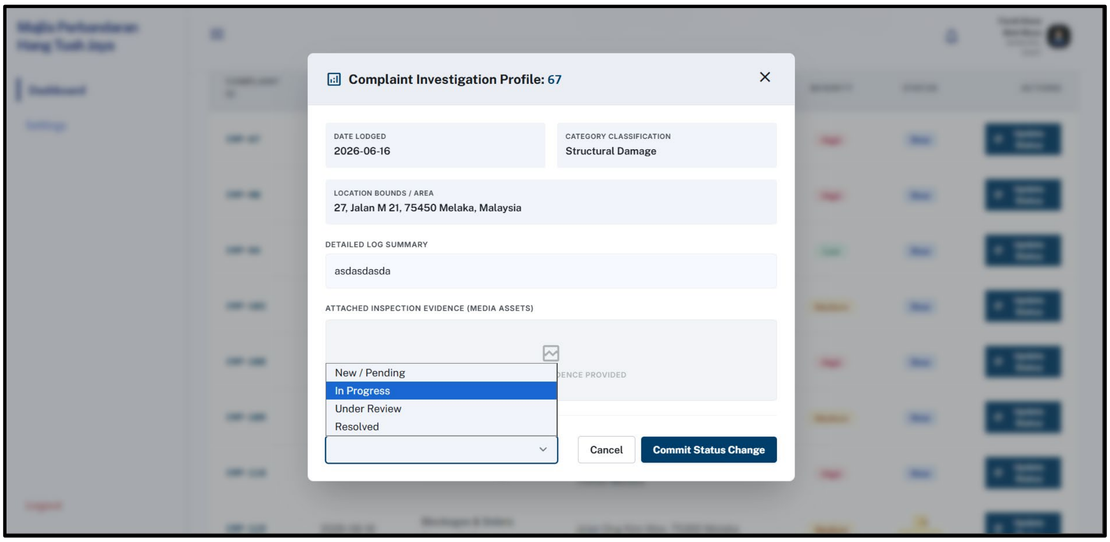

Figure 4.3.12 Staff Update Progress

Senior Manager positions have the authority to change the status of a report. Status reports will be changed after every update from the dispatch team.

- Output Design: Show how results or reports are displayed.

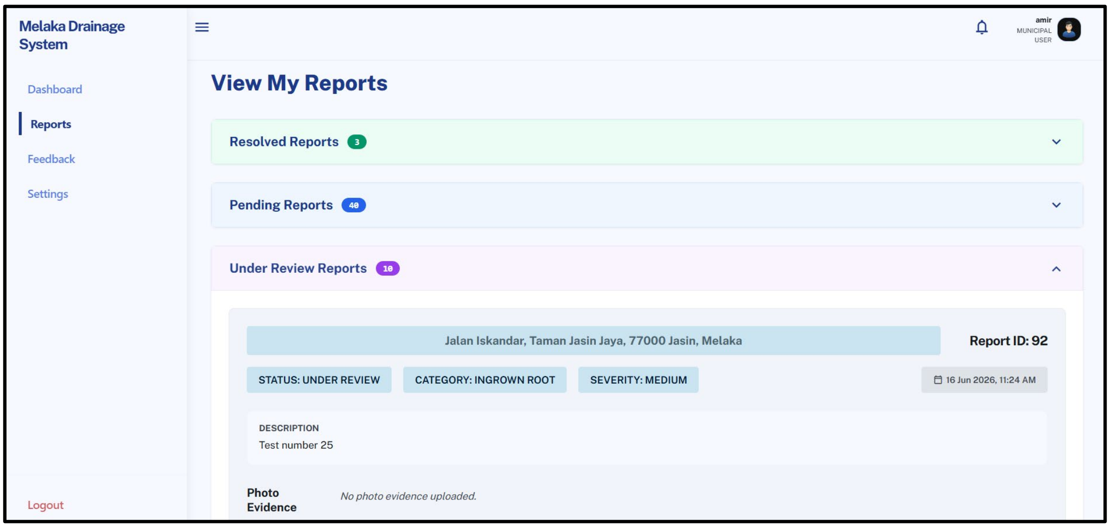

Figure 4.3.13 Citizen View Reports.

This page shows the report history of the current user. It is divided into the status of the reports.

Figure 4.3.14 Staff View Reports

All staff members can view reports and their details to prepare themselves before dispatch.

4.4 Conclusion

The chapter has covered the system architecture, database design, and supporting components that form the backbone of the drainage complaint management system. The three-tier architecture ensures clear separation of responsibilities, with the presentation layer handling user interaction, the application layer managing business logic, and the database layer maintaining secure and reliable storage. The database design built around normalized tables such as User, Staff, Municipality, Report, and Feedback, enforces integrity through foreign keys and constraints, while triggers automate rules like severity classification and municipality assignment.

Together, these design choices directly support the system's goals of efficiency, transparency, and accountability. The architecture enables scalable user interaction across citizens, staff, and administrators, while indexes and views optimize performance for reporting and monitoring. By aligning logical models with physical implementation, the system ensures accurate complaint handling, streamlined workflows, and actionable feedback, ultimately strengthening municipal service delivery.